In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
```

In [7]:
```
df = pd.read_csv("C:/Users/mestr/Downloads/archive (13)/sales_data_sample.csv",
df.head()
```

Out[7]:

	ORDERNUMBER	QUANTITYORDERED	PRICEEACH	ORDERLINENUMBER	SALES	OR
0	10107	30	95.70	2	2871.00	
1	10121	34	81.35	5	2765.90	
2	10134	41	94.74	2	3884.34	
3	10145	45	83.26	6	3746.70	
4	10159	49	100.00	14	5205.27	

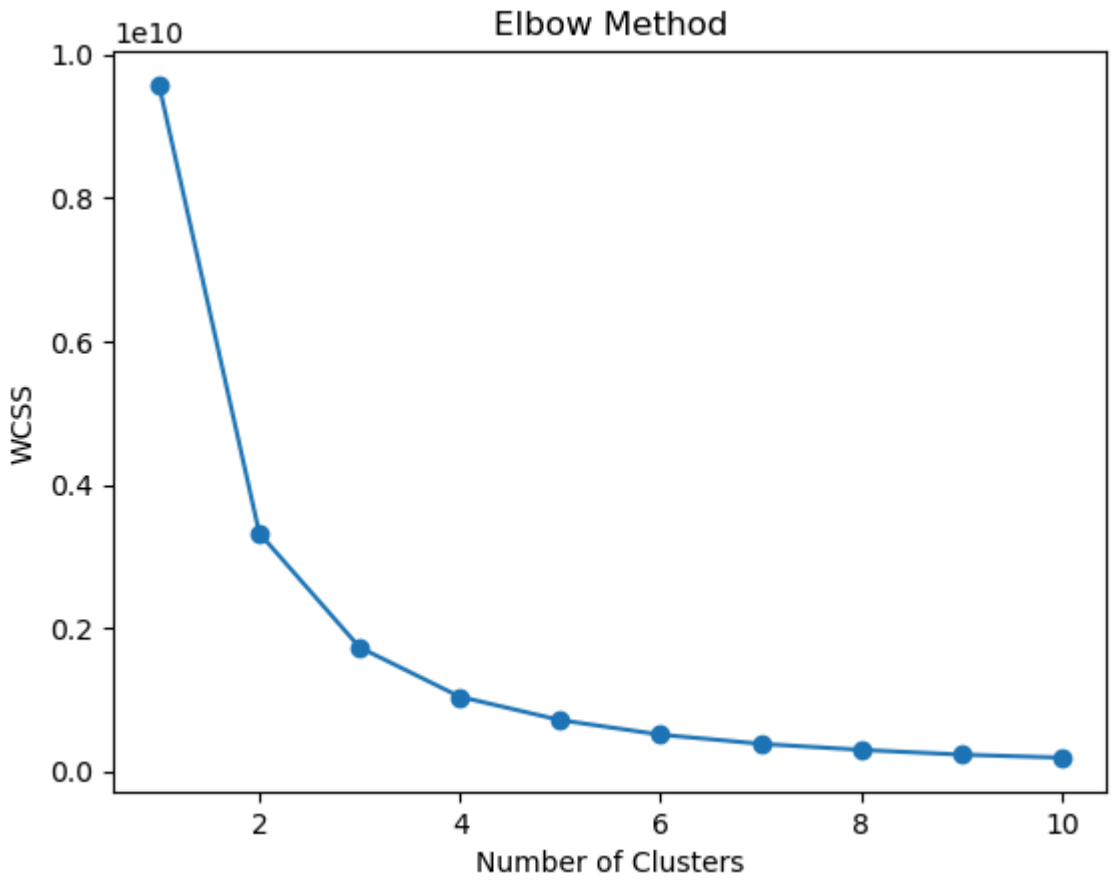
5 rows × 25 columns

In [8]:
```
X = df[['QUANTITYORDERED', 'PRICEEACH', 'SALES']]
X = X.dropna()
```

In [9]:
```
wcss = []

for i in range(1, 11):
    km = KMeans(n_clusters=i, random_state=42)
    km.fit(X)
    wcss.append(km.inertia_)
```

In [10]:
```
plt.plot(range(1, 11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```



In [11]:
```
km = KMeans(n_clusters=3, random_state=42)
y_pred = km.fit_predict(X)

print(y_pred)
```

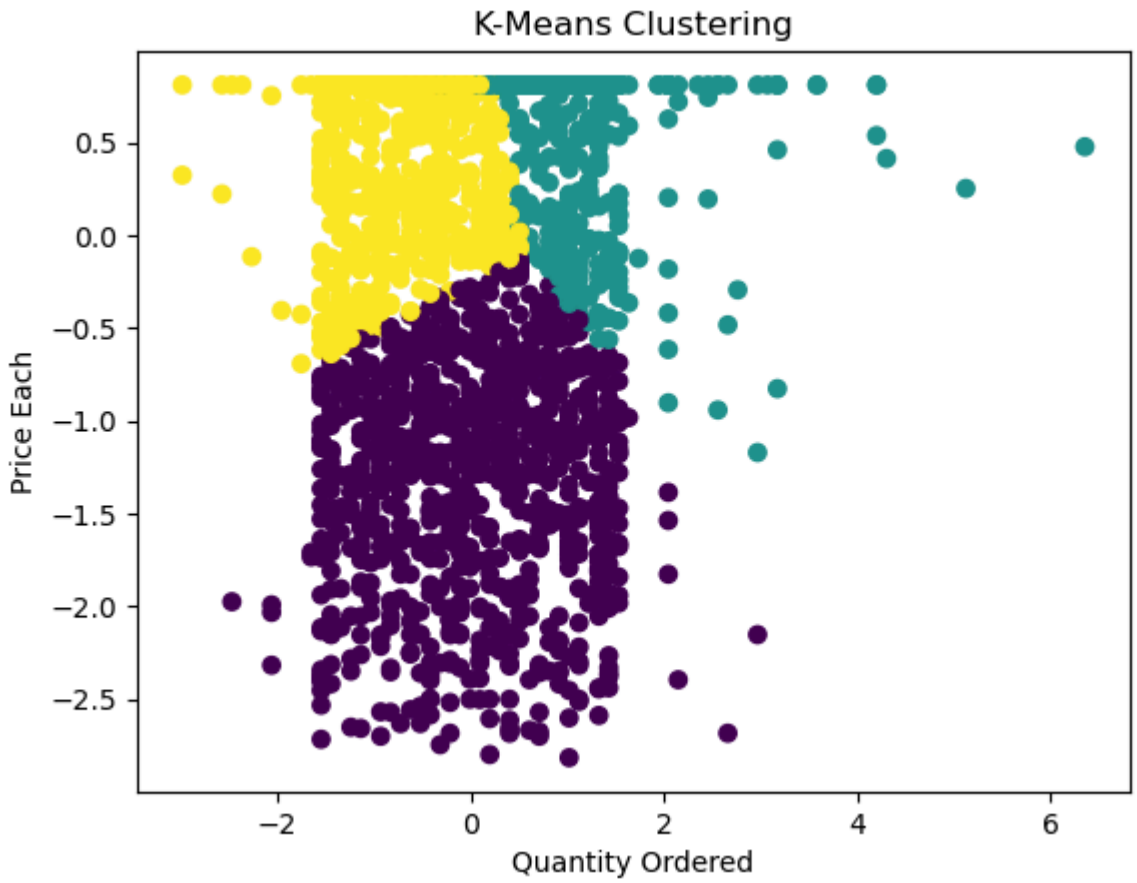
```
[0 0 1 ... 1 0 0]
```

In [14]:
```
sc = StandardScaler()
```

In [15]:
```
X = sc.fit_transform(df[['QUANTITYORDERED', 'PRICEEACH', 'SALES']].dropna())

km = KMeans(n_clusters=3, random_state=42)
y_pred = km.fit_predict(X)

plt.scatter(X[:, 0], X[:, 1], c=y_pred)
plt.xlabel("Quantity Ordered")
plt.ylabel("Price Each")
plt.title("K-Means Clustering")
plt.show()
```



In [16]:
```
print("Cluster Centers:")
print(km.cluster_centers_)
```

```
Cluster Centers:
[[ 0.00340709 -1.31519107 -0.83846133]
 [ 0.89846585  0.64278204  1.10107381]
 [-0.79188456  0.55612886 -0.2525054 ]]
```

In [ ]: